

Detailed Performance Summary For Qx_100to20_dTe_const_Tc_const_20C

Project: 30XW_China
Prepared By: Mikhail Terekhov

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AquaForce Greenspeed Water Cooled Screw Chiller



Unit Information

Tag Name: **Qx_100to20_dTe_const_Tc_const_20C**
Model Number: **30XW-VB459**
Condenser Type: **Water Cooled**
Nameplate Voltage: **400-3-50** V-Ph-Hz
Quantity: **3**
Manufacturing Source: **Shanghai, China**
Chiller Type: **Cooling**
Operating Mode: **Cooling**
Refrigerant: **R-515B**
Minimum Capacity: **20.00** %
Shipping Weight: **9421** kg
Operating Weight: **9327** kg
Unit Length: **4524** mm
Unit Width: **1745** mm
Unit Height: **2632** mm

Performance Information

Cooling Capacity: **1 170.** kW
Heating Capacity: **1 336** kW
Total Unit Power: **187.4** kW
Cooling Efficiency (COPR): **6.243** kW/kW
Heating Efficiency (COPH): **7.127** kW/kW

Acoustics Information

A-Weighted Sound Pressure Level: **81** dBA

Evaporator Information

Fluid Type: **Fresh Water**
Fouling Factor: **0.0180** (sqm-K)/kW
Leaving Temperature: **6.00** °C
Entering Temperature: **13.00** °C
Fluid Flow: **39.84** L/s
Pressure Drop: **19.7** kPa

Condenser Information

Fluid Type: **Propylene Glycol**
Brine Concentration: **45.0** %
Fouling Factor: **0.0440** (sqm-K)/kW
Leaving Temperature: **28.00** °C
Entering Temperature: **20.00** °C
Fluid Flow: **44.06** L/s
Pressure Drop: **31.3** kPa

Electrical Information

Unit Voltage: **400-3-50** V-Ph-Hz

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Standby Power:.....**0.05000** kW
Minimum Voltage:.....**360** Volts
Maximum Voltage:.....**440** Volts
Power Factor:.....**0.93**

Amps (Un)	Electrical Circuit 1	Electrical Circuit 2
Max Unit Current Draw (RLA)	464.0	---
Nominal Start Up Current (ICF)	294.0	---
Nominal Unit Current Draw (A)	294.0	---

Accessories and Installed Options

Opt. 282 EMC C2 Classification
Opt. 001A Integrated VFD

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Load Line

Unit Performance	**	***	(1)							
User-specified Cooling Capacity, kW		1000.0	900.0	800.0	700.0	600.0	500.0	400.0	300.0	200.0
Percent of User-specified Cooling Capacity, %	117	100.00	90.00	80.00	70.00	60.00	50.00	40.00	30.00	20.00
Percent Full Load Cooling Capacity, %	100.0	85.47	76.93	68.38	59.83	51.28	42.74	34.19	25.64	17.09
Percent of Full Load Power, %	100.0	83.54	76.19	59.99	50.22	44.43	36.12	27.44	18.45	11.79
Unloading Sequence	Default	Default	Default	Default	Default	Default	Default	Default	Default	Default
Cooling Capacity, kW	1 170.	1 000.	900.0	800.0	700.0	600.0	500.0	400.0	300.0	200.0
Heating Capacity, kW	1 336	1 138	1 025	898.8	782.4	672.7	558.7	444.4	329.6	219.8
Total Unit Power, kW	187.4	156.6	142.8	112.4	94.12	83.27	67.70	51.43	34.57	22.09
Nominal Unit Current Draw (RLA)	289.75	248.3	---	173.3	146.1	129.7	106.1	80.6	54.2	34.6
Cooling Efficiency kW/kW	6.243	6.387	6.303	7.116	7.438	7.205	7.386	7.777	8.678	9.054
Heating Efficiency, kW/kW	7.127	7.270	7.176	7.995	8.314	8.079	8.253	8.640	9.536	9.950
A-Weighted Sound Pressure Level, dBA	81	80	-	77	78	76	74	73	73	73
Evaporator Data										
Fluid Entering Temperature, °C	13.00	13.00	13.00	13.00	13.00	12.86	11.72	10.57	9.43	8.29
Fluid Leaving Temperature, °C	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00
Fluid Flow Rate, L/s	39.84	33.99	30.59	27.19	23.79	20.81	20.80	20.80	20.80	20.80
Pressure Drop, kPa	19.7	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Fouling Factor, (sqm-K)/kW	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180
Condenser Data										
Fluid Entering Temperature, °C	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00
Fluid Leaving Temperature, °C	28.00	28.00	27.24	26.34	25.52	24.75	23.94	23.13	22.32	21.55
Fluid Flow Rate, L/s	44.06	37.44	37.43	37.47	37.48	37.45	37.46	37.47	37.50	37.50
Pressure Drop, kPa	31.3	23.2	23.3	23.4	23.5	23.6	23.7	23.7	23.8	23.9
Fouling Factor, (sqm-K)/kW	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440

Note:

(**): Column displaying effective unit full load performances

(***): Column displays unit performance at the 'required capacity' and design conditions specified

(1) Compressor cycling is expected. Stated performances are average.

Sound pressure level was determined in accordance with AHRI Standard 575. Sound pressure level is the free field sound data (the unit is placed in free field over a reflecting plane).

Outside the scope of AHRI Water-Cooled Water-Chilling and Heat Pump Water-Heating Packages Certification Program or not optionally certified, but is rated in accordance with AHRI Standard 550/590 (I-P) and AHRI Standard 551/591 (SI).